

For both original trajectory as well as the rotated trajectory

1. The smaller values of the learning rate [more specifically learning rates of 0.05, 0.1, 0.2, 0.3, 0.4, 0.5] lead to successful convergence whereas the larger values [0.5, 1.0] did not lead to convergence.
2. Among converged executions, the smaller the learning rate, the more the number of iterations required for convergence. For example, for Rotated A, 0.05 learning rate required 531 iterations for convergence, 0.1 learning rate required 254 iterations for convergence and so on.
3. Among non-converged executions, we can not definitive say, if lesser the learning rate, lesser the no. of iterations required to figure out convergence is not attainable. In fact, larger learning rate of 1.0 required 22 iterations to bail out whereas learning rate of 0.5 took 3 iterations to bail out. This sort of indicates divergence and overshooting associated with larger learning rates.